
CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Glennie, Craig

PERSISTENT IDENTIFIER (PID): <https://orcid.org/0000-0003-1570-0889>

*POSITION TITLE: Professor

*ORGANIZATION AND LOCATION: The University of Houston, Houston, Texas, United States

Proposals and Active Projects

*Proposal/Active Project Title: STV Incubation Study Team Technology Co-Lead
2025-2029

*Status of Support: Current

Proposal/Award Number:

*Source of Support: NASA

*Primary Place of Performance: Houston, TX

*Proposal/Active Project Start Date: (MM/YYYY): 10/2025

*Proposal/Active Project End Date: (MM/YYYY): 09/2029

*Total Anticipated Proposal/Project Amount: \$227,479

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1
2027	1
2028	1
2029	1

***Overall Objectives:**

Technology Co-Lead for Decadal Survey Incubation Study for development of a high-resolution satellite based 3D observing system for measuring surface topography and vegetation (STV).

***Statement of Potential Overlap:**

None

***Proposal/Active Project Title:** Enhancing Graduate Geodesy at The University of Houston

***Status of Support:** Current

Proposal/Award Number:

***Source of Support:** NGA

***Primary Place of Performance:** University of Houston

***Proposal/Active Project Start Date: (MM/YYYY):** 06/2025

***Proposal/Active Project End Date: (MM/YYYY):** 05/2030

***Total Anticipated Proposal/Project Amount:** \$2,132,116

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	0.5
2026	0.5
2027	0.5
2028	0.5
2029	0.5

***Overall Objectives:**

The objective of this proposal is to establish a graduate research student pipeline for the MSc and PhD programs in Geosensing Systems Engineering and Sciences (GSES) at the Cullen College of Engineering (CCE) at the University of Houston (UH); focusing on geodetic practice through coursework, research and outreach activities.

***Statement of Potential Overlap:**

Students in this current project, if it continues, will be working in tandem with the proposed students. These synergies will allow more momentum for a graduate geodesy pipeline of professionals.

***Proposal/Active Project Title:** Facility Support to Renew Operation of the National Center for Airborne Laser Mapping (NCALM)

***Status of Support:** Current

Proposal/Award Number: 2324629

***Source of Support:** NSF

***Primary Place of Performance:** University of Houston

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2024

***Proposal/Active Project End Date: (MM/YYYY):** 08/2027

***Total Anticipated Proposal/Project Amount:** \$1,062,651

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	1
2026	1
2027	0.1

***Overall Objectives:**

Since NCALM inception in 2003 it has had three objectives: 1) To train and educate graduate students to meet the needs of private industry, government agencies, and academic institutions. 2) To advance the state of the art in airborne lidar technology and its application by the research community, and 3) To provide research quality airborne lidar data to the scientific community.

***Statement of Potential Overlap:**

None

***Proposal/Active Project Title:** Analysis Capabilities for Competition, Crisis, and Combat (AC4) (PI)

***Status of Support:** Current

Proposal/Award Number: W911QX23D0009

***Source of Support:** Army Devcom Analysis Center

***Primary Place of Performance:** University of Houston

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2023

***Proposal/Active Project End Date: (MM/YYYY):** 09/2028

***Total Anticipated Proposal/Project Amount:** \$63,500,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2023	2
2024	1
2025	1

Year	Person Months
2026	1
2027	1

***Overall Objectives:**

Modeling and Simulation for Devcom Analysis Center. Providing frameworks and assessment schemes for Army needs during competition and crisis.

***Statement of Potential Overlap:**

None

***Proposal/Active Project Title:** Collaborative Research: Geomorphological Change in in the McMurdo Dry Valleys: AI-Based Stream Mapping and Multi-Temporal Topography to investigate fluvial thermokarst changes

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** NSF

***Primary Place of Performance:** University of Houston

***Proposal/Active Project Start Date: (MM/YYYY):** 01/2027

***Proposal/Active Project End Date: (MM/YYYY):** 12/2029

***Total Anticipated Proposal/Project Amount:** \$325,303

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	0.75
2028	0.75
2029	0.75

***Overall Objectives:**

The McMurdo Dry Valleys (MDV) are the largest ice-free region in Antarctica and are increasingly recognized as a landscape on the threshold of climate-driven change which could lead to reorganizing of biogeochemical systems. We have developed a new, scalable framework for detecting stream channel change by combining deep-learning stream-boundary segmentation, co-registration of multi-source DEMs, and uncertainty-aware topographic differencing. Using lidar-based digital elevation models we predict the location and extent of streams in the MDV. From multi-year photogrammetric digital elevation models, the elevation change within the channels are calculated

and net rates of stream incision and deposition calculated. We hypothesize that the largest geomorphic changes occur where meltwater pathways intersect ice-cemented or ice-rich permafrost in the Coastal Thaw Zone, whereas drier terrain in inland valleys will show less sediment mobility and weaker erosion. The project will produce valley-wide maps of geomorphic sensitivity to fluvial erosion and predictive models for where incision, bank failure, aggradation, widening, and lateral migration are most likely to accelerate under continued warming. This work relies on existing data and no field work is requested.

***Statement of Potential Overlap:**

This proposed project.

***Proposal/Active Project Title:** Integrated Multi-Sensor Monitoring of Arctic Coastal Hazards

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** ESTCP

***Primary Place of Performance:** Houston, TX

***Proposal/Active Project Start Date: (MM/YYYY):** 01/2027

***Proposal/Active Project End Date: (MM/YYYY):** 12/2028

***Total Anticipated Proposal/Project Amount:** \$1,951,616

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	0.5
2028	0.5

***Overall Objectives:**

Establish a baseline coastal monitoring framework for the Utqiagvik and Prudhoe Bay coastlines by integrating multiple sensors, including airborne lidar, satellite imagery, and prior coastal surveys, to quantify shoreline position and coastal morphology.

***Statement of Potential Overlap:**

None

***Proposal/Active Project Title:** Advancing Geodesy Workforce Development Through an Education-to-Careers Pipeline at the University of Houston

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** NGA

***Primary Place of Performance:** University of Houston

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2026

***Proposal/Active Project End Date: (MM/YYYY):** 08/2031

***Total Anticipated Proposal/Project Amount:** \$2,429,129

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	0.5
2028	0.5
2029	0.5
2030	0.5
2031	0.5

***Overall Objectives:**

This project will establish a sustainable workforce pipeline of students trained in mission-relevant geodesy applications. We will provide structured entry-level research experiences and advanced graduate education to develop professionals uniquely qualified to support NGA's workforce needs.

***Statement of Potential Overlap:**

None

***Proposal/Active Project Title:** Gulf Artificial Intelligence Learning Center for Severe Weather and Infrastructure Systems (GAIL-CSWIS)

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** DOE

***Primary Place of Performance:** University of Houston

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2026

***Proposal/Active Project End Date: (MM/YYYY):** 08/2029

***Total Anticipated Proposal/Project Amount:** \$5,000,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	1
2028	1
2029	1

***Overall Objectives:**

This proposal responds to DOE's call for proposals and establishes the Gulf Artificial Intelligence Learning Center for Severe Weather and Infrastructure Systems (GAIL-CSWIS). CSWIS will serve as a multidisciplinary and innovation hub dedicated to advancing the understanding, prediction, and mitigation of severe weather-related hazards that threaten energy, power, and water systems, and other critical infrastructure across the Gulf Region.

***Statement of Potential Overlap:**

This proposal

***Proposal/Active Project Title:** Creating a NASA Surface Topography and Vegetation (STV) Novel Observing System (NOS)

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** NASA

***Primary Place of Performance:** Houston, TX

***Proposal/Active Project Start Date: (MM/YYYY):** 06/2026

***Proposal/Active Project End Date: (MM/YYYY):** 05/2029

***Total Anticipated Proposal/Project Amount:** \$4,206,117

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.25
2027	0.25
2028	0.25

***Overall Objectives:**

To develop a novel observing system (NOS) demonstration to enable the implementation of NASA's Surface Topography and Vegetation (STV) Targeted Observable.

***Statement of Potential Overlap:**

None

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§ 287, 1001, 1031 and 31 U.S.C. §§ 3729-3733 and 3802.

Certified by Glennie, Craig in SciENCv on 2026-07-14 12:35:57